L05: Support Vector Machines

CSci 574 Machine Learning (Géron) Ch. 5

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Large Margin Classification

When two classes can be separated by a line (hyperplane).

- They are **linearly separable**

SVM classifier fits the widest possible street.

- This is called **large margin classification**

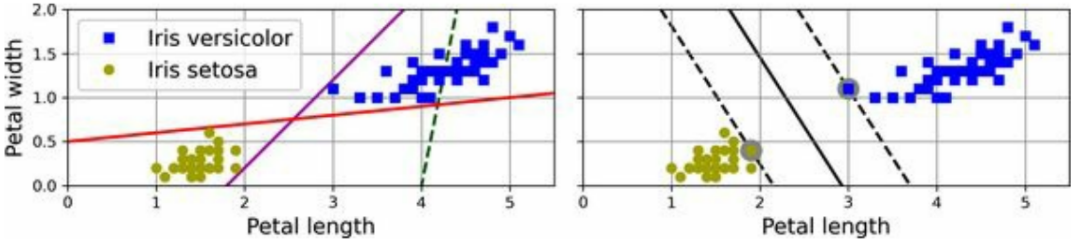

Figure 1: Large margin classification (Géron, 2023, pg.287)

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Linear Support Vector Machine

- Notice that adding more training instances "off the street" will not affect the decision boundary at all.
  - It is fully determined (or "supported") by the instances located on the edge of the street
  - These instances are called the **support vectors**
- A Support Vector Machine (SVM) is a machine learning model capable of performing linear or nonlinear classification and regression.
- It works by identifying support vectors to determine the largest margin decision boundary

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SVMs are Sensitive to Scaling

In the left plot vertical scale is much larger than the horizontal scale, gives a almost horizontal large margin with these support vectors.

In the right plot after standard feature scaling, the decision boundary had a much different slope.

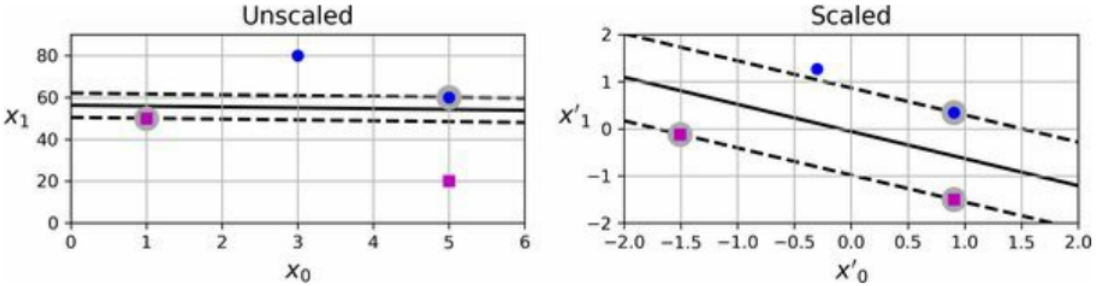

Figure 2: SVM sensitivity to feature scales. (Géron, 2023, pg.288)

Soft Margin Classification

If we strictly require all instances off street and on correct side of margin, this is called **hard margin classification**.

Only possible when data are linearly separably.

- Figure 3 left, impossible to find hard margin, not linearly separable.

Also sensitive to outliers.

- Figure 3 right, decision boundary ends up very different if there is an outlier here, might not generalize.

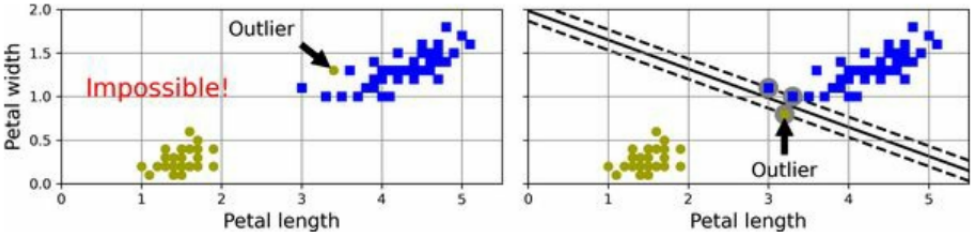

Figure 3: Hard margin sensitivity to outliers. (Géron, 2023, pg.288)

Soft Margin Classification

- To avoid both, find balance between keeping street as large as possible and limiting the **margin violations**
  - This is **soft margin classification**
- SVM use regularization hyperparameter `C`
  - When `C` is low, large margin, more margin violations allowed, but if too low can cause underfitting.
  - When `C` is high, small margins, less margin violations allowed, but can overfit and be sensitive to outliers

SVM C Regularization

If your SVM model is overfitting, you can regularize it by reducing C, leading to softer margins which will tend to be better at generalizing performance to unseen data.

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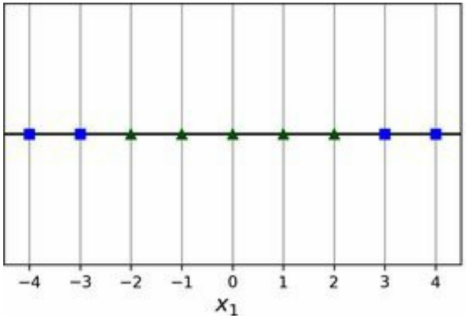
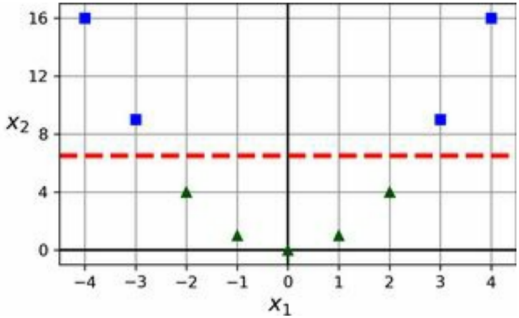
Nonlinear SVM Classification

- Many datasets are not even close to being linearly separable.
- One approach is to add more polynomial feature combinations of the original features.
- This can result in transforming data into a linearly separable dataset.



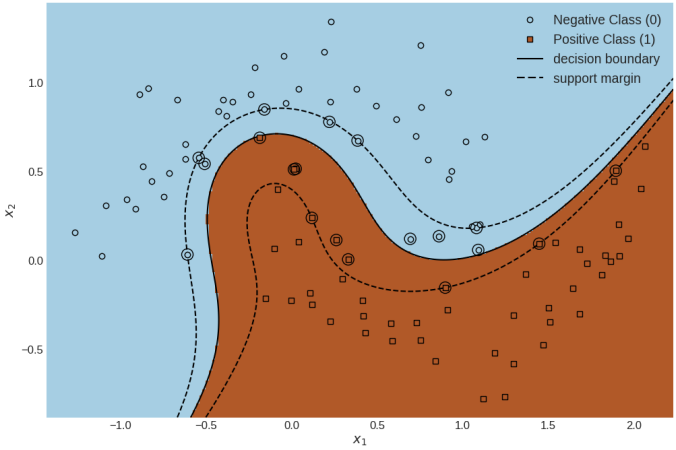
Single feature x_1 is not linearly separable.



But if we add new polynomial feature $x_2 = (x_1)^2$, resulting 2D dataset is linearly separable.

Adding Nonlinear Features by Hand

As we have done previously for logistic regression, add in nonlinear polynomial features by hand.




```python
# adding degree 3 polynomials combinations
# to 2 feature nonlinear dataset
# makes resulting nonlinear features
# linearly separable
X, y = make_moons(n_samples=100, noise=0.15)

polynomial_svm_clf = Pipeline([
    ("poly_features", PolynomialFeatures(degree=3)),
    ("scaler", StandardScaler()),
    ("clf", SVC(kernel='linear', C=10, max_iter=10000))
])

polynomial_svm_clf.fit(X, y)
```

Figure 4: Linear SVM classifier using polynomial features.

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Polynomial Kernel, the **Kernel Trick**

- Adding polynomial features can work with all sorts of ML (not just SVMs)
  - But at low polynomial degree, cannot deal with complex datasets
  - And at high polynomial degree, creates huge (combinatorial explosion) number of features (slow).
- When using SVM, can use the **kernel trick**
  - Can give same result as if added many polynomial features, without the computational costs of actually adding them.
- The kernel trick is built-in to the `scikit-learn` `SVC` class.

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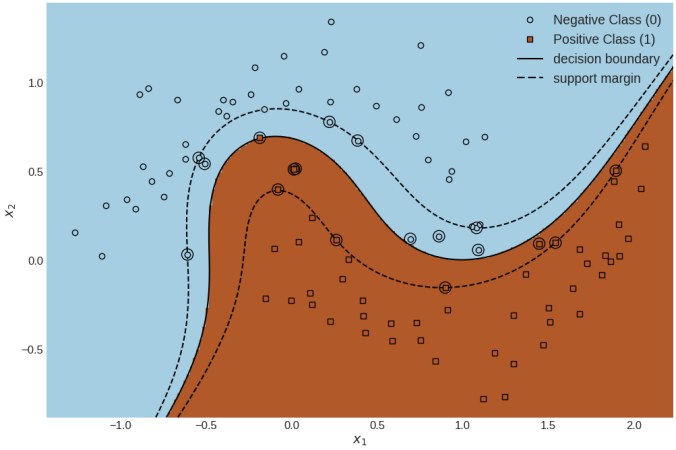
SVC Polynomial Kernel




```python
# using a degree 3 polynomial kernel for a SVC,
# this implementation will be much faster for large
# polynomial degrees / large datasets
from sklearn.svm import SVC

poly_kernel_svm_clf = Pipeline([
    ("scaler", StandardScaler()),
    ("clf", SVC(kernel="poly", degree=3, coef0=1, C=5))
])

poly_kernel_svm_clf.fit(X, y)
```

Figure 5: SVC Using degree 3 polynomial kernel (implemented using the kernel trick).

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SVM Kernel Regularization

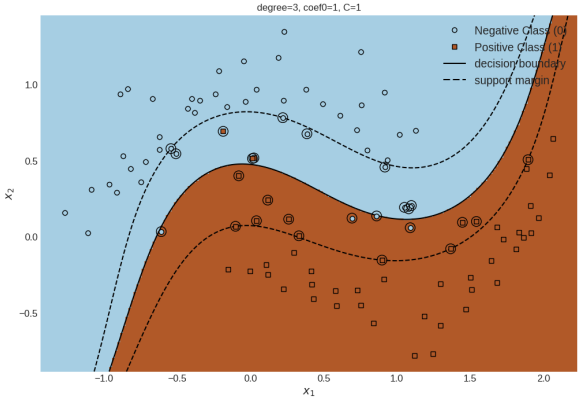

Figure 6: SVC Using degree 3 polynomial kernel, large regularization, if need to underfit.

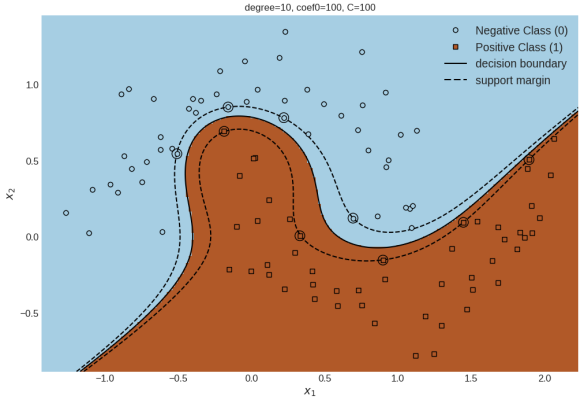

Figure 7: SVC Using degree 10 polynomial kernel, small regularization, if need to overfit.

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Bibliography

Géron, A. (2023). *Hands-on machine learning with scikit-learn, keras and tensorflow* (third). O'Reilly Media, Inc.

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